COSMIC Installation Guide

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1 Introduction

COSMIC (Compact Object Synthesis and Monte Carlo Investigation Code) is a rapid binary population synthesis suite with a special focus on generating compact binary populations. This guide will walk you through the installation process.

2 Prerequisites

Before installing COSMIC, you need to have certain prerequisites installed on your system:

2.1 Installing gfortran

Since COSMIC requires compilation of Fortran code, you'll need a gfortran installation. The installation method depends on your operating system:

2.1.1 For MacOS Users

You can install gfortran using Homebrew:

```
brew install gcc
```

Note for Apple Silicon Users

If you're using an Apple Silicon (ARM) processor, verify your Python architecture:

```
python -c "import platform; print(platform.architecture())"
```

If the output shows `(’64bit’, ’x86_64’)`, install the ARM version:

```
brew install python@3.10
```

2.1.2 For Unix/Linux Users

Install gfortran using your package manager:

```
sudo apt-get install gfortran
```

2.1.3 For Windows Users

Windows is not directly supported due to issues with the gfortran compiler and libraries. However, you can use Windows Subsystem for Linux (WSL):

1. Install WSL by following Microsoft's guide at: `https://docs.microsoft.com/en-us/windows/wsl/install`
2. Follow the Unix/Linux instructions within WSL

3 Installing COSMIC

We recommend using conda to create a dedicated environment for COSMIC. This ensures all dependencies are installed correctly.

3.1 Setting up the Conda Environment

First, install Miniconda if you haven't already from: `https://docs.conda.io/en/latest/miniconda.html`

3.1.1 For MacOS and Unix/Linux

```
conda create -n cosmic numpy h5py python=3.10
source activate cosmic
pip install cosmic-popsynth
```

4 Using Jupyter with COSMIC

To use COSMIC with Jupyter notebooks, you need to install Jupyter in the COSMIC environment:

4.1 Using conda

```
conda install jupyter ipython
```

4.2 Using pip

```
pip install jupyter ipython
```

Important Note

Always use Jupyter and IPython from within the COSMIC conda environment. Using the global instance may cause issues.

5 Troubleshooting

- **MacOS Users**: If you encounter issues, try reinstalling gfortran and then reinstall COSMIC
- **Linux Users**: Make sure your gfortran installation is up to date
- **WSL Users**: Ensure you're using WSL 2 and have installed all required Linux components

6 Next Steps

After installation, you can verify your installation by running a simple test:

```
from cosmic.sample.initialbinarytable import InitialBinaryTable
from cosmic.evolve import Evolve
```

If no errors appear, your installation is successful!